TSA Airport Claims: An Operational and Financial Analysis

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Abstract

This report investigates Transportation Security Administration (TSA) insurance claims data to identify operational vulnerabilities and financial trends across U.S. airports. By cleaning and analyzing a multi-variable dataset, this study identifies the most frequent types and sites of claims, calculates approval rates, and visualizes the geographic concentration of claim activity. Key findings reveal that "Checked Baggage" remains the primary site of property loss. While claim volumes fluctuate, the financial impact remains an operational concern for the agency, requiring improvements in high-volume hubs.

Introduction

The TSA is in charge of securing the nation's transportation systems, a mission involving handling millions of passenger property daily. Despite established protocols, property damage and loss do occur, leading travelers to file insurance claims. High claim volumes create financial liability for the agency and can erode public trust over time. This analysis offers TSA leadership a data-driven foundation for resource allocation and process improvement, with a focus on identifying where property is most at risk.

Analytic Goals

The primary objective of this analysis is to uncover actionable insights from historical claims data. Specific goals include:

- Determining which claim types and airport sites are most prone to incidents.
- Establishing a baseline for "typical" claim amounts and the overall agency approval rate.
- Mapping the geographic distribution of claims to identify high-risk airport hubs.

- Analyzing temporal trends to evaluate whether total payouts are increasing or decreasing over time.

Data

The analysis utilized three primary data sources:

1. **TSA Claims Dataset (tsa_claims2.csv):** Contains record-level data on claim types, sites, amounts, and outcomes.
2. **Global Airport Database:** Provided IATA codes and decimal coordinates for spatial mapping.
3. **U.S. State Shapefiles:** Provided the geographic boundaries for the visualization.

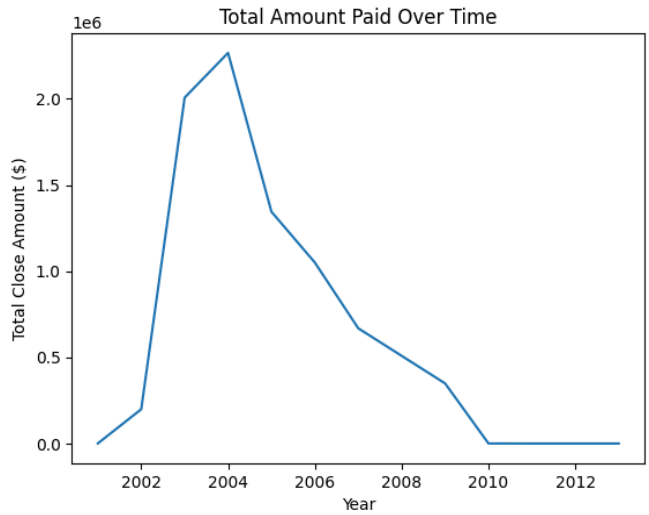
Data Limitations & Assumptions: The data required extensive cleaning due to inconsistent currency formatting (symbols and commas) and missing values. It is assumed that "Incident Date" is the most accurate metric for temporal analysis. Records with dates before 2002 were excluded as data entry anomalies.

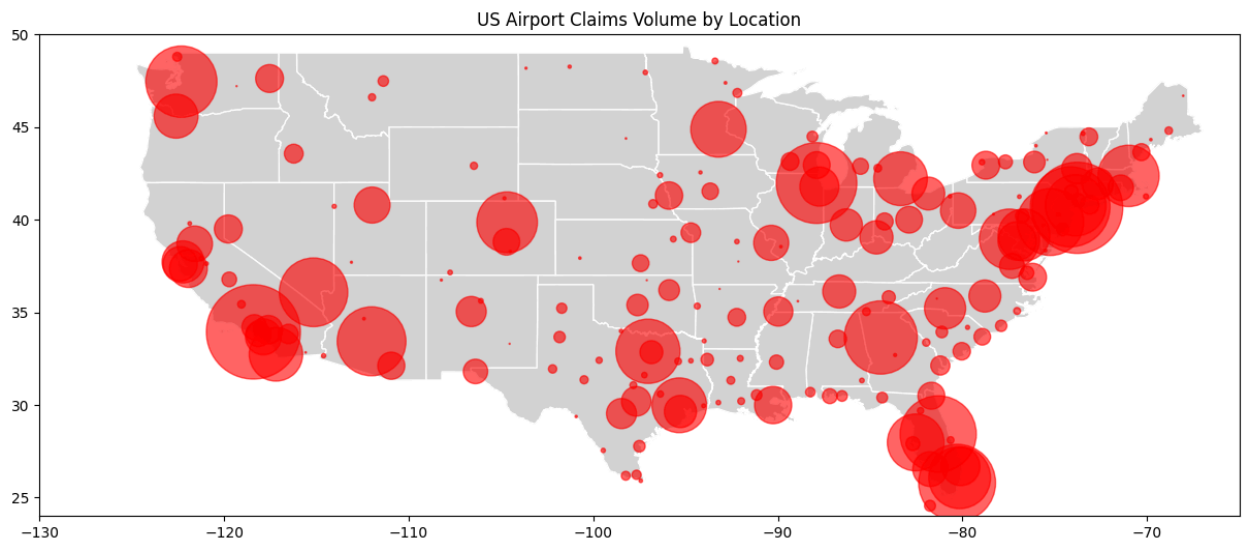
Methods

The analysis was conducted using Python and several specialized libraries:

- **Data Preparation:** Used `Pandas` to strip currency symbols and convert objects to floats for mathematical operations. Dates were standardized into `datetime` objects. Missing categorical data in "Claim Site" and "Claim Type" were labeled as "Unknown" to maintain dataset integrity.
- **Statistical Analysis:** Calculated the median for claim amounts to avoid the skewing effect of high-value outliers. Approval rates were calculated by isolating "Approved" and "Settled" statuses.
- **Visualization:** Created a time-series line chart to visualize financial trends and utilized `GeoPandas` to generate a proportional symbol map. Marker sizes were scaled based on claim counts per airport.

Findings





- **Claim Commonality:** "Passenger Property Loss" and "Property Damage" emerged as the most frequent claim types.
- **High-Risk Locations:** The "Checked Baggage" site has consistently reported the highest frequency of claims, followed by "Checkpoint" areas.
- **Financial Metrics:** The analysis determined the typical (median) claim amount to be $100.00, while the overall approval rate was 33.87%.
- **Geographic Distribution:** Major international hubs, including LAX, JFK, and ORD, showed the highest concentration of claims, as evidenced by the scaled markers in the geographic visualization.
- **Temporal Trends:** Total close amounts were aggregated by year to determine the trajectory of TSA payouts over the studied period.

Discussion and Conclusion

The geographic and operational concentration of claims within "Checked Baggage" indicates that property remains most vulnerable while out of the passenger's direct supervision. Furthermore, the **33.87% approval rate** reflects a stringent adjudication framework, likely due to documentation gaps or liability limitations. These findings are vital for the agency as it navigates the balance between passenger sentiment and fiscal accountability.

Recommendations:

1. **Targeted Training:** Focus baggage handler training and surveillance enhancements at the top five airports identified in the geographic map.
2. **Passenger Awareness:** Since property loss is the leading claim type, increasing awareness regarding high-value item packing could reduce TSA liability.
3. **Efficiency Audit:** Review the "Checkpoint" procedures at major hubs to identify if specific screening technologies correlate with higher damage claims.

References

1. McCoy, S. (2024). *Murach's Python for Data Science* (2nd ed.). Mike Murach & Associates.
2. Transportation Security Administration (TSA) Claims Dataset.
3. Global Airport Database.